ARNAU NECHES VILÀ

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EDUCATION

The University of Sydney – Sydney, Australia Aug. 2025 – Jun. 2026
Exchange Student, Mathematics & Computer Science | GPA: 93/100 | Top 5%
Selected coursework: Natural Language Processing, Deep Learning, Operating Systems Internals, Differential Geometry, Nonlinear ODEs with Applications, Complex Analysis, Financial Derivatives

Universidad de Sevilla – Seville, Spain Sep. 2022 – Expected May 2027
Double Degree in Mathematics & Computer Science | GPA: 9.3/10 | Highest honours in 78+ ECTS
Selected coursework: Real & Complex Analysis, Fourier Analysis, Functional Analysis, Probability Theory, Statistical Inference, Mathematical Modelling, Abstract Algebra, Mathematical Programming, Algorithms & Data Structures, Artificial Intelligence, Computer and Network Architecture

RESEARCH EXPERIENCE

MINERVA AI Lab, Universidad de Sevilla – Research Assistant Jul. 2025 – Present

- Currently researching vision foundation models for cancer detection in MRI scans, evaluating their ability to extract clinically useful representations for downstream diagnostic tasks.
- Developed and evaluated two new feature-selection methods designed to better capture interactions between features; led the research, implementation, experimental design, and evaluation.
- First author of "Lookahead and Dynamic Beam Search Variants of FOCI for Conditional-Dependence Feature Selection," published in Lecture Notes in Artificial Intelligence (LNAI), CAEPIA 2026.
- Co-authored "Benchmarking Chronos-2 as a Foundation Model for Traffic Forecasting," published in Lecture Notes in Artificial Intelligence (LNAI), CAEPIA 2026.

PROFESSIONAL EXPERIENCE

Welyto – Lead Software Engineer Apr. 2026 – Aug. 2026
Startup | Conversational AI device for older adults

- Took an AI-powered conversational device for older adults from development to deployment with 50 paying beta families using physical devices, supporting concurrent conversations, persistent user memories and preferences, and family-to-device messaging.
- Designed and built the backend and device-cloud architecture, spanning AI orchestration, cloud infrastructure, real-time WebSocket communication, tool execution, memory storage, and the family-facing application.
- Built the platform using Python, FastAPI, PostgreSQL, WebSockets, Google Cloud, and Vertex AI, and contributed to the software architecture of the next-generation product.

SELECTED PROJECTS

Accessible Activity Data Platform – University of Sydney Aug. 2025 – Oct. 2025
Project Manager & Developer | 7-person team | Client: Institute of Musculoskeletal Health

- Led development of a platform helping people with disabilities discover suitable physical activities, coordinating the technical work and client relationship.
- Built an AI-powered data collection pipeline that discovered activity providers online and transformed unstructured web content into structured data; developed the database and web interface and integrated the Gemini API and Google Maps API.

Distributed Warehouse Infrastructure – Universidad de Sevilla Oct. 2026 – Nov. 2026
Distributed Systems & DevOps Course Project | 6 physical nodes

- Built and deployed a distributed warehouse-order processing system across six physical nodes using K3s, RabbitMQ, Helm, and Ansible.
- Implemented reliable message processing, autoscaling, and observability, and tested recovery under Kubernetes- and RabbitMQ-level failures.

LEADERSHIP & ACTIVITIES

Competitive Programming Club – Universidad de Sevilla 2024 – Present
Events & Institutional Relations

- Active competitive programmer; qualified with my team for the national Ada Byron programming contest in both 2024 and 2025.
- Organize academic and industry events and manage external relations, building relationships with 5 companies for sponsorships, talks, and collaborations.

Sputnik / Los 300 2025

- Selected for Los 300 after completing Sputnik, a selective programme bringing together young people from southern Spain around technology, entrepreneurship, innovation, leadership, and social impact.

SKILLS

Programming: Python, C, Java, JavaScript, SQL, MATLAB
ML & Data: PyTorch, scikit-learn, Pandas, NumPy
Software: FastAPI, React, PostgreSQL, SQLAlchemy, WebSockets, REST APIs
Cloud & Infrastructure: Google Cloud Platform, Docker, Kubernetes/K3s, Helm, Ansible, RabbitMQ, Prometheus, Grafana, Git, Linux
Languages: Spanish, Catalan, English